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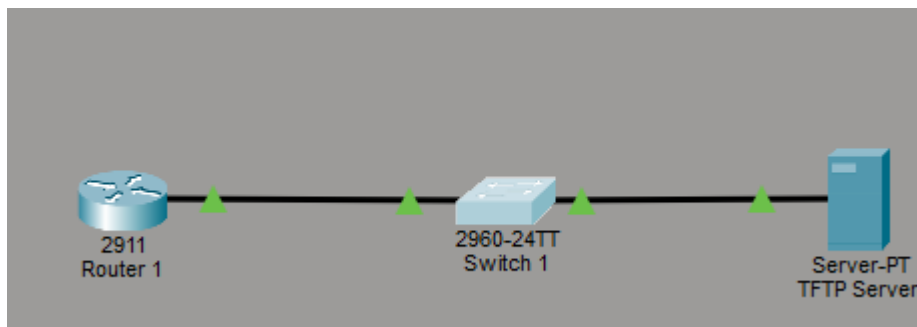
# NET512 WEEK 6 PRAC

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Lerato Modise



## Screenshot of the topology



## Configuring router and show interface brief

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 10.10.10.1 255.0.0.0
Router(config-if)#ip address 10.10.10.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to
up

Router(config-if)#exist
^
% Invalid input detected at '^' marker.

Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  10.10.10.1      YES manual up          up
GigabitEthernet0/1  unassigned      YES unset  administratively down down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
Router#
```

## Give a router IP address

```
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.1.10 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief


| Interface          | IP-Address   | OK? | Method | Status                | Protocol |
|--------------------|--------------|-----|--------|-----------------------|----------|
| GigabitEthernet0/0 | 192.168.1.10 | YES | manual | up                    | up       |
| GigabitEthernet0/1 | unassigned   | YES | unset  | administratively down | down     |
| GigabitEthernet0/2 | unassigned   | YES | unset  | administratively down | down     |
| Vlan1              | unassigned   | YES | unset  | administratively down | down     |


Router#
```

## Showing the successful ping

```
Router#ping 192.168.1.100

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.100, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/3/14 ms
```

## Configuration backup

```
Router# copy running-config tftp:
Address or name of remote host []? 192.168.1.100
Destination filename [Router-config]?

Writing running-config.....
%Error opening tftp://192.168.1.100/Router-config (Timed out)
Router#
```

## IOS transfer

```
System flash directory:
File Length Name/status
 3 33591768 c2900-universalk9-mz.SPA.151-4.M4.bin
 2 28282 sigdef-category.xml
 1 227537 sigdef-default.xml
[33847587 bytes used, 221896413 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)

Router#copy flash:tftp:
% Incomplete command.
Router#copy flash: tftp:
Source filename []? c2900-universalk9-mz.SPA.151-4.M4.bin
Address or name of remote host []? 192.168.1.100
Destination filename [c2900-universalk9-mz.SPA.151-4.M4.bin]?

Writing c2900-universalk9-mz.SPA.151-4.M4.bin.....
%Error opening tftp://192.168.1.100/c2900-universalk9-mz.SPA.151-4.M4.bin (Timed out)
Router#
```

## Restore normal boot

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#config-register 0x2102
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]? reload
%Error copying nvram:reload (Invalid argument)
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#reload
Proceed with reload? [confirm]ySystem Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE
(fcl)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO2911/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 72/-1(On-board/DIMM0) bit mode with ECC disabled

Readonly ROMMON initialized

program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test

Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x3bcd3d8
Self decompressing the image :
##### [OK]
Smart Init is enabled
smart init is sizing iomem
          TYPE      MEMORY_REQ
Onboard devices &
```

## Check the configuration

```
Router>enable
Password:
Password:
Password:
Router#erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete
%SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
Router#reload
Proceed with reload? [confirm]n
Router#show startup-config
startup-config is not present
Router#show running-config
Building configuration...

Current configuration : 756 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
enable secret 5 $1$mERr$1oJ053xwDS1EWhg2PBRFF0
!
!
!
!
!
ip cef
no ipv6 cef
--More--
```

```
interface GigabitEthernet0/0
 ip address 192.168.1.10 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
!
ip flow-export version 9
!
!
--More-- |
```

## Test the TFTP server

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.1.10 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 192.168.1.100

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.100, timeout is 2 seconds:
!!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/3/12 ms
```

---

```
Router#show running-config
Building configuration...

Current configuration : 756 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
enable secret 5 $1$mERr$loJ053xwDS1EWhg2PBRFF0
!
!
```

---

```
!
!
interface GigabitEthernet0/0
ip address 192.168.1.10 255.255.255.0
duplex auto
speed auto
!
!
```

---

## IOS image from TFTP

```
Router#show flash:

System flash directory:
File Length Name/status
 3 33591768 c2900-universalk9-mz.SPA.151-4.M4.bin
 2 28282 sigdef-category.xml
 1 227537 sigdef-default.xml
[33847587 bytes used, 221896413 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)

Router#copy tftp: flash:
Address or name of remote host []? 192.168.1.100
Source filename []? c2900-universalk9-mz.SPA.151-4.M4.bin
Destination filename [c2900-universalk9-mz.SPA.151-4.M4.bin]?
```

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